

Chapter 7 — Introduction to bias and fairness

Earlier chapters established how datasets are collected, labeled, cleaned, and explored

Learning objectives

- Identify policy and governance pressures that make fairness an operational requirement

Dataset bias and fairness

- Dataset bias is a systematic error or imbalance that skews training or evaluation data
- Fairness is the operational goal of selecting, measuring
- Fairness is therefore a measurable design constraint

Trust and accountability

- Trust declines when people cannot determine whether a recommendation
- Accountability requires named owners, documented data and metrics, mechanisms for review
- Deployment, and monitoring

Example 7.4 — GDPR & ECOA Policies

- Example 7.4 — hands-on module
- Example 7.4 shows that fairness has a policy context
- European data protection rules place duties around consequential automated decisions
- A technically impressive model can still create legal, financial
- View files: ``modules/chapter7/example4/``

Example 7.6 — Gender Shades Project

- Example 7.6 — hands-on module
- Example 7.6 records subgroup evidence from commercial facial analysis systems
- With underrepresentation in training data identified as an important cause
- The practical lesson is to report disaggregated errors before deployment rather than
- View files: ``modules/chapter7/example6/``

Example 7.9 — OECD AI Principles

- Example 7.9 — hands-on module
- Example 7.9 places fairness within a broader governance framework
- The OECD principles emphasize transparent, fair, inclusive
- Corrective action
- View files: ``modules/chapter7/example9/``

Fairness across the lifecycle

- Bias can enter through collection
- It can then be amplified during model training and deployment
- Fairness work must therefore span the lifecycle

Takeaways

- Dataset bias describes systematic skew
- Average performance can conceal serious subgroup failures
- Trust depends on transparency and consistent treatment
- Policy principles and legal duties make these concerns part of routine engineering

Next

- Complete the quiz for this part
- The next clip moves from the general definition to a practical taxonomy